Convert the following values to decimal using the double-dabble algorithm. Decimal values should be expressed as fractional values. **Write the value at the current step of the operation in the space provided under each binary number**. **[5 points]**

1 0 0 1 . 0 1 1 0 1 = $9\frac{13}{32}$

1 1 0 0 . 1 1 0 1 0 = $12\frac{26}{32}$

<u>Reformulation (formulaic)</u>

Reformulate by rearranging, applying appropriate trigonometric identities, and simplifying. **Show your work. [10 points]**

$1-\sqrt{\cos x}$

$$\left(1-\sqrt{\cos x}\right)\left(\frac{1+\sqrt{\cos x}}{1+\sqrt{\cos x}}\right)=\frac{1-\cos x}{1+\sqrt{\cos x}}$$

$$=\left(\frac{1-\cos x}{1+\sqrt{\cos x}}\right)\left(\frac{1+\cos x}{1+\cos x}\right)$$

$$=\frac{\sin^2 x}{\left(1+\sqrt{\cos x}\right)(1+\cos x)}$$

Reformulate using the Mean Value Theorem. **Show your work. [10 points]**

$$\frac{1}{x^2+2x+1}-\frac{1}{x^2}$$

$f(x)=\frac{1}{x^2}$ $\qquad$ $f'(x)=-\frac{2}{x^3}$

$\theta=\frac{2x+1}{2}$ $\qquad$ $f(b)-f(a)=(b-a)f'(\theta)$

$$(x+1-x)\left(-\frac{2}{\theta^3}\right)=-\frac{2}{\left(\frac{2x+1}{2}\right)^3}$$

Reformulation (numerical)

Reformulate the following numerical problem to increase precision. Calculate the relative error before and after reformulation. Use 3 significant digits for each operation (use normal rounding rules). **[20 points]**

$\cos(\pi+0.061)+1$ $\qquad$ exact: $.186\times10^{-2}$

$\cos\pi\cos\epsilon-\sin\pi\sin\epsilon+1$ $\qquad$ computed: $.002$

$-\cos\epsilon+1$ $\qquad$ reformulated: $.186\times10^{-2}$

$$(1-\cos\epsilon)\left(\frac{1+\cos\epsilon}{1+\cos\epsilon}\right)=\frac{1-\cos^2\epsilon}{1+\cos\epsilon}=\frac{\sin^2\epsilon}{1+\cos\epsilon}$$

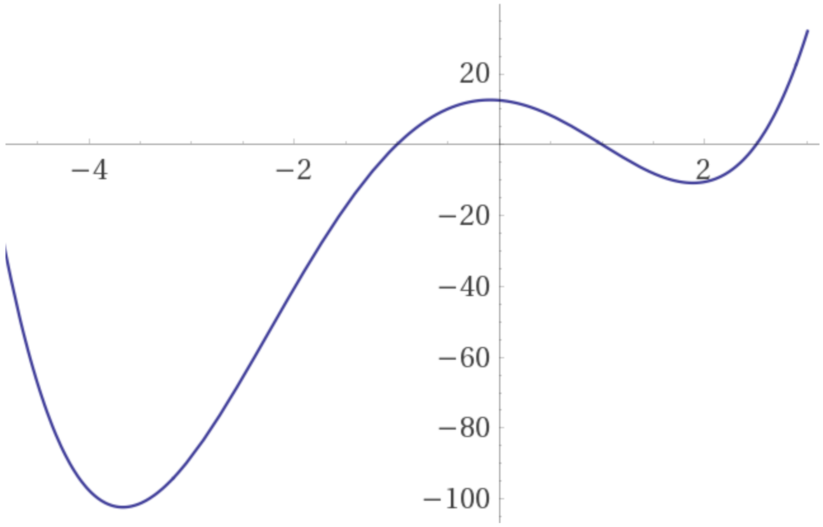
$$\frac{\sin^2 0.061}{1+\cos 0.061}=\frac{(0.061)^2}{1+0.998}=\frac{0.00372}{2}=0.00186$$

error before: $\left|\frac{.002-.00186}{.00186}\right|=8.1\%$

error after: $\left|\frac{.00186-.00186}{.00186}\right|=0\%$

Roots of Equations – Bisection and Newton's Method

The following is a plot of the polynomial $p(x) = (x+5)(x-2.5)(x+1)(x-1)$:



- Calculate the zero using **3 iterations** of the bisection method given the interval $[-4, 0.75]$. Fill the table below, and include the **relative** error for each step. Keep 3 significant digits. **[10 points]**

a	$f(a)$	b	$f(b)$	c	$f(c)$	E_r
-4	-97.5	0.75	4.402	-1.625	-22.84	62.5%
-1.625	-22.84	0.75	4.402	-0.4375	10.84	56.3%
-1.625	-22.84	-0.4375	10.84	-1.031		3.1%

- Calculate the root using **2 iterations** of Newton's method given $x_0 = 2$ given the expanded polynomial:

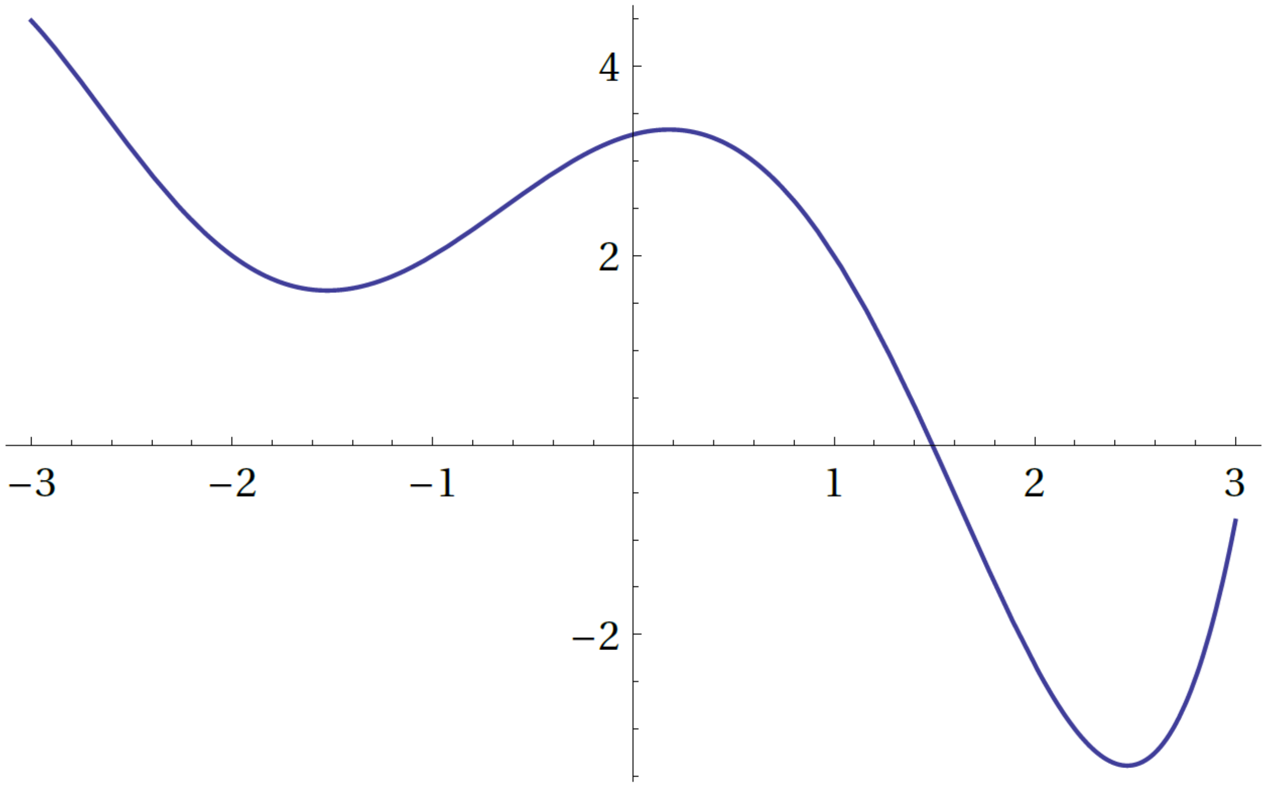
$$x^4 + 2.5x^3 - 13.5x^2 - 2.5x + 12.5$$

Fill the table below, and include the **relative** error at each step. Keep 3 significant digits. **[10 points]**

x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	E_r
2	-10.5	5.5	3.91	56%
3.91	179.5	245.7	3.175	27%

Roots of Equations – Method of False Position

Draw and label the first **three** steps of the method of false position to find the root at $x = 1.5$ given the interval $[-3, 3]$ in the following function. **[5 points]**



Algorithm Design – Horner's Algorithm

Use synthetic division to compute $\sin x \cos x$ for $x = 0.1$ using a 5[th]-order Taylor series expansion about zero. Show how you derive the polynomial formulation **[10 points]**.

$$f(x) = \sin x \cos x \qquad f(0) = 0$$

$$f'(x) = \cos^2 x - \sin^2 x \qquad f'(0) = 1$$

$$f''(x) = -4 \sin x \cos x \qquad f''(0) = 0$$

$$f'''(x) = -4(\cos^2 x - \sin^2 x) \qquad f'''(0) = -4$$

$$f^{(4)} = 16 \sin x \cos x \qquad f^{(4)} = 0$$

$$f^{(5)} = 16(\cos^2 x - \sin^2 x) \qquad f^{(5)} = 16$$

$$p(x) = x - \frac{2}{3}x^3 + \frac{2}{15}x^5$$

Show this calculation using synthetic division with 4 significant digits. [15 points]

0.1 |	.1333	0	−.6667	0	1	0
		.01333	.001333	-.06654	-.006654	.09933
	.1333	.01333	-.6654	-.06654	.9933	.09933

Algorithm Design – Floating Point

[5 points] What is the largest decimal value you can represent using a 4-bit mantissa (m) and a 3-bit exponent (c). Assume that an implied 1 is used for the mantissa and the exponent has a bias of 4:

$$(-1)^s \times 1.m \times 2^{c-4}$$

$$(1.1111)_2 \times 2^{(111)_2-4} = (1.1111)_2 \times 2^3 = (1111.1)_2 = 15\tfrac{1}{2} = 15.5$$

[5 points] What error can we expect for calculations that have values in this range?

The smallest representable value is $\mathbf{(1.0000)_2 \times 2^3 = 8}$

The next representable value is $\mathbf{(1.0001)_2 \times 2^3 = 8\tfrac{1}{2} = 8.5}$

The maximum step size is $\mathbf{8 - 8.5 = .5}$, therefore the maximum error will be $\pm\frac{\mathbf{0.5}}{\mathbf{2}} = \pm\mathbf{0.25}$